NANDHAKUMAR G

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PROFESSIONAL SUMMARY

Mechanical Engineering student with hands-on experience in reverse engineering, product design, industrial systems, CAD-driven workflows and technical documentation. Adept at converting engineering concepts into structured, implementation-ready information while collaborating across multidisciplinary environments. Strong foundation in mechanical systems, digital tools and process-oriented problem solving with a proven ability to quickly learn and adapt to new technologies.

SELECTED HIGHLIGHTS

- Experience across reverse engineering, product design, industrial systems and manufacturing environments.
- Strong ability to convert engineering concepts into structured implementation-ready information.
- Exposure to multidisciplinary engineering workflows from design through validation and documentation.
- Hands-on experience with both physical systems and digital engineering platforms.

TECHNICAL CAPABILITIES

Technical Information & Documentation

- Technical Documentation
- Engineering Specifications
- Technical Reporting
- Drawing Interpretation
- Information Organization

Engineering Analysis & Validation

- Reverse Engineering
- Assembly Validation
- Design Optimization
- Process Analysis
- Tolerance Analysis

Digital Engineering & Systems

- CAD Workflows
- Systems Thinking
- Workflow Optimization
- Data Interpretation
- Cross-functional Collaboration

Software & Digital Tools

Fusion 360 | Siemens NX | SolidWorks | AutoCAD | Python | MATLAB | COMSOL | Blender | HTML | CSS

PROFESSIONAL EXPERIENCE

Reverse Engineering Intern **Femlogic Technologies Pvt. Ltd.** June 2026

- Captured high-accuracy dimensional data of legacy mechanical systems using precision metrology tools to establish reliable baseline technical documentation.
- Executed digital engineering workflows to reconstruct physical geometries into parametric 3D CAD models, resolving design deviations and optimizing structures for modern assembly systems.
- Performed clearance checks, fit validation, and tolerance stack analysis to verify cross-functional interface compatibility.

Industrial Engineering Trainee **Steel Authority of India Ltd. (SAIL) – Salem Steel Plant** December 2025

- Supported engineering operations through drawing interpretation of heavy steel rolling mills and mechanical system schematics.
- Managed technical documentation by organizing preventative maintenance protocols, operational checklists, and safety manuals.
- Collaborated with maintenance teams on process analysis and data interpretation of equipment logs to trace component wear histories.

Product Design Intern **Femlogic Technologies Pvt. Ltd.** June 2025 – July 2025

- Participated in CAD workflows and design validation of structural seating assemblies for high-reliability transit train systems.
- Reviewed structural interface specifications and fasteners selection, ensuring compliance with manufacturing (DFM) requirements.
- Drafted technical specifications and conducted tolerance analysis on sheet metal parts to satisfy quality validation reviews.

Production Engineering Intern **K.M. Knitwear Pvt. Ltd.** July 2024 – August 2024

- Conducted process analysis on floor packaging layouts, developing workflow optimization schemes that successfully increased output rates by 10%.
- Established structured shipping layouts and dimension templates, improving information organization to minimize transport-related parts damage.
- Provided engineering support during routine diagnostics and maintenance cycles, tracing mechanical issues to reduce machinery downtime by 15%.

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MAJOR ENGINEERING PROJECTS

Project EcoFloat (Niral Hackathon Regional Winner)
ROS · IoT · Fusion 360 · Arduino · Sensor Fusion

- Co-developed a solar-powered, dual-function autonomous robotic catamaran engineered to simultaneously clear floating debris and analyze water chemical toxicity parameters.
- Implemented a real-time toxicity sensing and mapping framework integrating ROS node communications and IoT data streams, converting field telemetry into structured data sets.
- Designed the twin-hull mechanical system structural geometry and sensor mounting interfaces in CAD, validating center of gravity and hydrodynamics clearance constraints.

Regenerative Braking System Simulation
MATLAB Simulink · BMS · Powertrain Modeling · Control Engineering

- Simulated an active regenerative braking control loop for EV powertrains in MATLAB Simulink, establishing mathematical models to map brushless motor kinetics and battery recovery rates.
- Executed data interpretation of transient current profiles during deceleration phases to optimize energy recovery, extending simulated battery range parameters by up to 20%.
- Documented system design validation reports detailing energy flow diagrams, safety limits, and thermal stress parameters under variable friction-braking loads.

Advanced Aquatic Monitoring Platform
Hyperspectral Imaging · Sensor Fusion · Optical Gimbals · Additive Manufacturing

- Designed a stabilized floating vessel equipped with a high-resolution optical spectrometer gimbal for real-time remote environmental mapping of lake water quality indicators.
- Processed and analyzed hyperspectral spectral datasets using custom scripts to perform chlorophyll concentration mapping and microplastic polymer detection.
- Executed digital engineering workflows to optimize structural gimbals and chassis components, validating designs for additive manufacturing limits to reduce wave disturbance interference.

Hemispherical CPV Solar Tracking System
COMSOL Multiphysics · Solar Tracking Math · Fusion 360 · MATLAB

- Designed a Concentrator Photovoltaic (CPV) tracking dome with dual-axis actuators driven by solar vector calculations to maximize daily energy absorption.
- Validated structural integrity, fluid-structure wind loads, and thermal characteristics of the compound panel shell using COMSOL Multiphysics finite element analysis (FEA).
- Created detailed engineering drawings, assembly drawings, and technical documentation specifications detailing clearance stack-ups and mechanical tolerances.

EDUCATION

Bachelor of Mechanical Engineering St. Joseph's College of Engineering, Chennai, TN — 2023 – Present

Higher Secondary Education (Class XII) Shri Swamy Matriculation Higher Secondary School, Salem, TN — 2022 – 2023

Secondary School Education (Class X) Shri Swamy Matriculation Higher Secondary School, Salem, TN — 2020 – 2021

CERTIFICATIONS

Autodesk Certified Fusion Designer
Professional CAD Design Credential

Certified Solidworks Designer
Mechanical Modeling & Validation Spec

Certified Siemens NX CAD Designer
Advanced Parametric Modeling Specialist

AutoCAD for 2D Modelling
Technical Drafting & Detailing Standard

Certified in Python Advanced
Computational Data Structures & Integration

Certified Frontend Developer
Core Interface Architect (HTML/CSS/JS)

Blender for 3D Modelling
Complex Surface Mesh Optimization

C++ Programming
Object-Oriented Software & Logic Design

ADDITIONAL INFORMATION

Languages

English – Fluent
Tamil – Native

Professional Strengths

Structured Technical Communication · Analytical Thinking · Fast Learning Ability · Attention to Detail · Engineering Collaboration · Digital Tool Adaptability